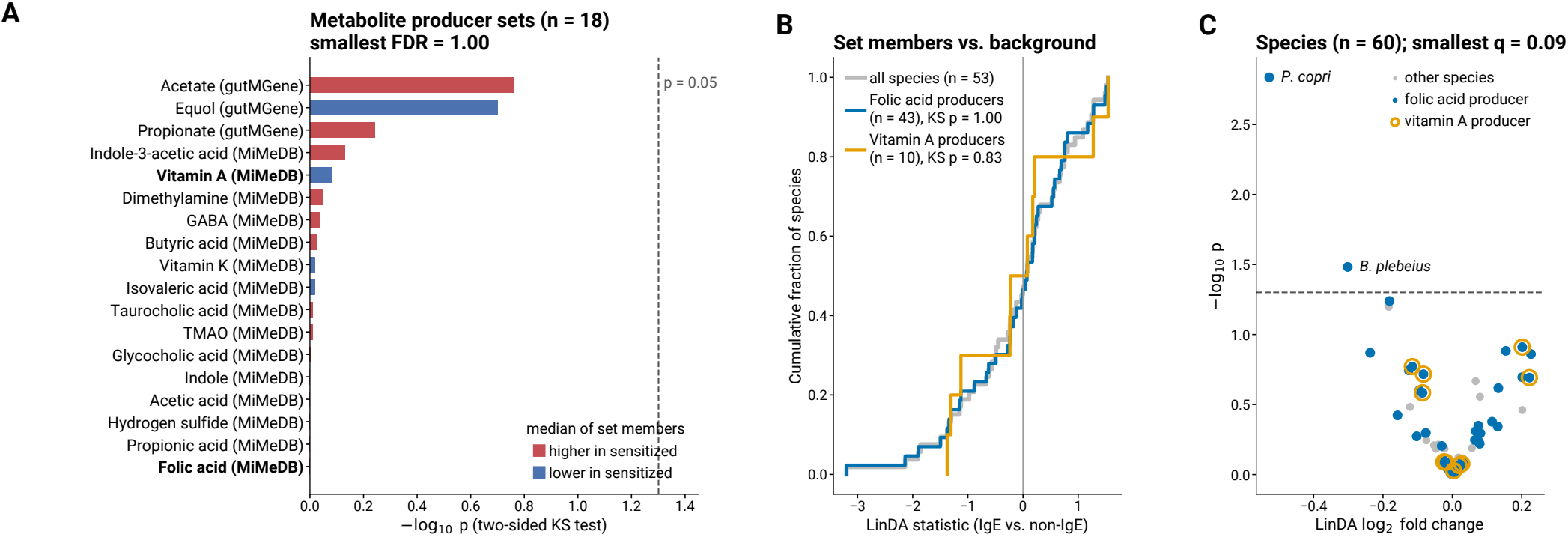
A
Metabolite producer sets (n = 18)
smallest FDR = 1.00
Acetate (gutMGene)
Equol (gutMGene)
Propionate (gutMGene)
Indole-3-acetic acid (MiMeDB)
Vitamin A (MiMeDB)
Dimethylamine (MiMeDB)
GABA (MiMeDB)
Butyric acid (MiMeDB)
Vitamin K (MiMeDB)
Isovaleric acid (MiMeDB)
Taurocholic acid (MiMeDB)
TMAO (MiMeDB)
Glycocholic acid (MiMeDB)
Indole (MiMeDB)
Acetic acid (MiMeDB)
Hydrogen sulfide (MiMeDB)
Propionic acid (MiMeDB)
Folic acid (MiMeDB)
p = 0.05
median of set members
higher in sensitized
lower in sensitized
0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4
−log10 p (two-sided KS test)
B
Set members vs. background
all species (n = 53)
Folic acid producers
(n = 43), KS p = 1.00
Vitamin A producers
(n = 10), KS p = 0.83
Cumulative fraction of species
0.0 0.2 0.4 0.6 0.8 1.0
−3 −2 −1 0 1
LinDA statistic (IgE vs. non-IgE)
C
Species (n = 60); smallest q = 0.09
P. copri
B. plebeius
other species
folic acid producer
vitamin A producer
−log10 p
0.0 0.5 1.0 1.5 2.0 2.5
−0.4 −0.2 0.0 0.2
LinDA log2 fold change